

Name: _____ Date: _____ Period: _____

2.5 Practice
Absolute Value Equations

Evaluate each expression if $x = -1$, $y = 3$, and $z = -4$.

1. $16 - |2z + 1|$

2. $|x - y| + 4$

3. $|-3y + z| - x$

4. $3|z - x| + |2 - y|$

Solve each equation. Then graph the solution set.

5. $-9|-z + 5| = 0$

6. $-|7 - 2r| + 11 = -8$

7. $13|-2t + 1| - 10 = 159$

8. $4|g - 9| + 7 = 67$

Write an equation involving absolute value for each graph.

13. FITNESS Taisha uses the elliptical cross-trainer at the gym. Her general goal is to burn 300 Calories per workout, but she varies by as much as 20 Calories from this amount on any given day. Write and solve an equation to find the maximum and minimum number of Calories Taisha burns on the cross-trainer.

Solutions: _____

Midpoint: _____

Distance from midpoint: _____

Equation: _____

14. TEMPERATURE A thermometer is guaranteed to give a temperature no more than 1.5°F from the actual temperature. If the thermometer reads 82°F , **write and solve** an equation to find the maximum and minimum.

Solutions: _____

Midpoint: _____

Distance from midpoint: _____

Equation: _____